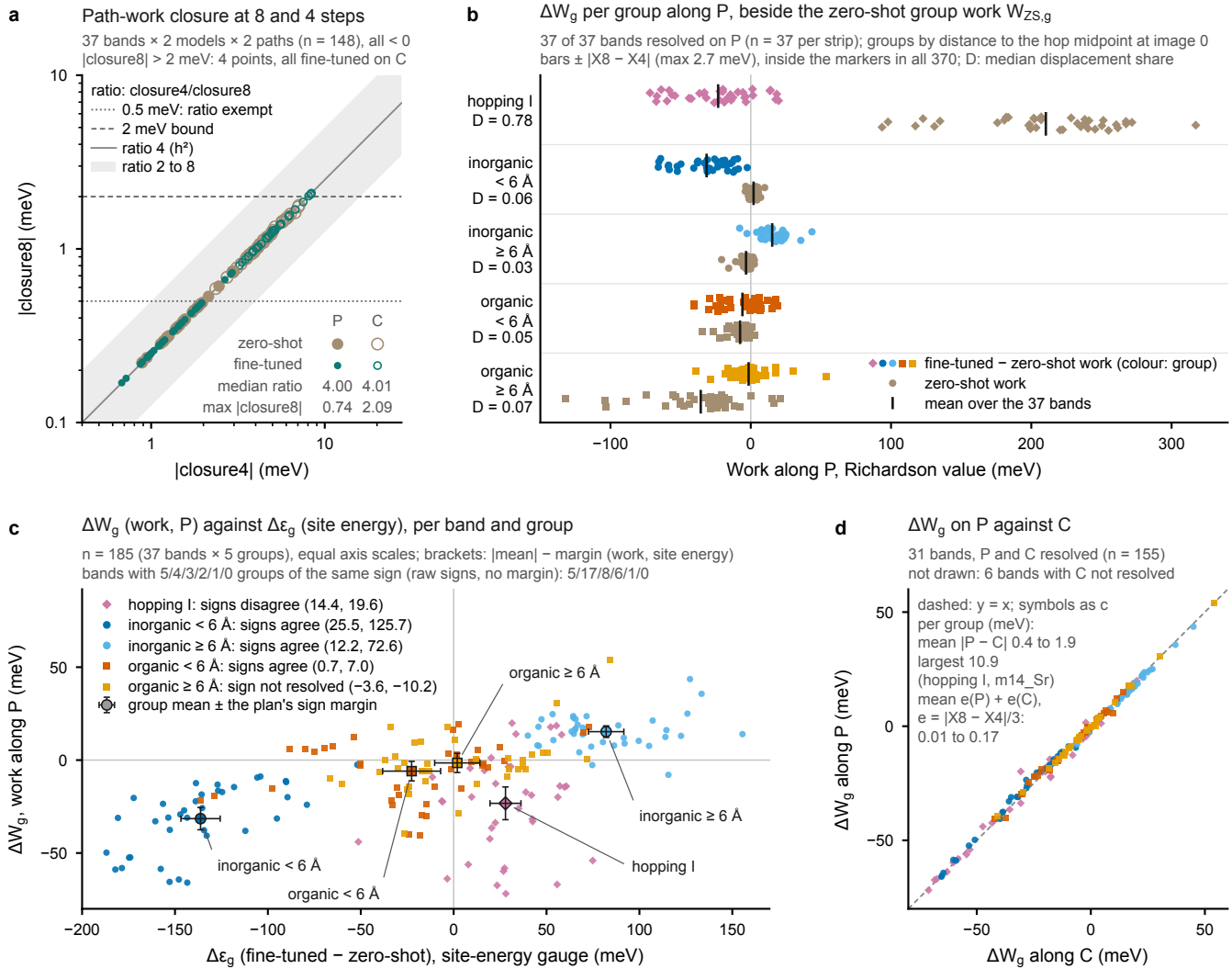

a
Path-work closure at 8 and 4 steps
37 bands × 2 models × 2 paths (n = 148), all < 0
|closure8| > 2 meV: 4 points, all fine-tuned on C
ratio: closure4/closure8
0.5 meV: ratio exempt
2 meV bound
ratio 4 (h²)
ratio 2 to 8
P C
zero-shot
fine-tuned
median ratio 4.00 4.01
max |closure8| 0.74 2.09
|closure8| (meV)
|closure4| (meV)
b
ΔW_g per group along P, beside the zero-shot group work W_ZS,g
37 of 37 bands resolved on P (n = 37 per strip); groups by distance to the hop midpoint at image 0
bars ± |X8 − X4| (max 2.7 meV), inside the markers in all 370; D: median displacement share
hopping I
D = 0.78
inorganic
< 6 Å
D = 0.06
inorganic
≥ 6 Å
D = 0.03
organic
< 6 Å
D = 0.05
organic
≥ 6 Å
D = 0.07
fine-tuned − zero-shot work (colour: group)
zero-shot work
mean over the 37 bands
Work along P, Richardson value (meV)
c
ΔW_g (work, P) against Δε_g (site energy), per band and group
n = 185 (37 bands × 5 groups), equal axis scales; brackets: |mean| − margin (work, site energy)
bands with 5/4/3/2/1/0 groups of the same sign (raw signs, no margin): 5/17/8/6/1/0
hopping I: signs disagree (14.4, 19.6)
inorganic < 6 Å: signs agree (25.5, 125.7)
inorganic ≥ 6 Å: signs agree (12.2, 72.6)
organic < 6 Å: signs agree (0.7, 7.0)
organic ≥ 6 Å: sign not resolved (−3.6, −10.2)
group mean ± the plan's sign margin
organic ≥ 6 Å
inorganic ≥ 6 Å
hopping I
inorganic < 6 Å
organic < 6 Å
ΔW_g, work along P (meV)
Δε_g (fine-tuned − zero-shot), site-energy gauge (meV)
d
ΔW_g on P against C
31 bands, P and C resolved (n = 155)
not drawn: 6 bands with C not resolved
dashed: y = x; symbols as c
per group (meV):
mean |P − C| 0.4 to 1.9
largest 10.9
(hopping I, m14_Sr)
mean e(P) + e(C),
e = |X8 − X4|/3:
0.01 to 0.17
ΔW_g along P (meV)
ΔW_g along C (meV)